Cristian F. López R.

(he/him/his)
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Postdoctoral Associate
Department of Applied Mathematics
University of Colorado Boulder

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RESEARCH INTERESTS

Weak form Scientific Machine Learning, Data-driven modeling and identification of dynamical systems, Model predictive control, Theoretical and experimental nonlinear dynamics and vibrations, Signal processing, Rotating machinery fault diagnosis.

EDUCATION

2021 – 2025	Ph.D. in Mechanical Engineering and Applied Mechanics University of Nebraska – Lincoln, Lincoln, Nebraska, USA Dissertation: "Data-driven Methods for Signal Analysis and Nonlinear System Identification" M.S. in Mechanical Engineering and Applied Mechanics Advisor: Keegan J. Moore.
2015 – 2017	M.Eng. in Thermal Power Engineering Zhejiang University, Hangzhou, Zhejiang, China Thesis: "Bearing Fault Diagnosis based on Improved Stochastic Resonance Method" Advisor: Wei Zhong.
2008 – 2013	B.S. in Mechanical Engineering Escuela Politécnica Nacional, Quito, Pichincha, Ecuador Thesis: "The Design of a Heat Exchanger for the Cupola Furnace" Advisor: Fausto Oviedo.

PROFESSIONAL EMPLOYMENT

2025 – date	**Postdoctoral Associate**, Department of Applied Mathematics, University of Colorado Boulder, Boulder, Colorado, USA. Advisors: David Bortz, Stephen Becker.
2022 – 2025	**Graduate Research Assistant – Fulbright Fellow**, Department of Mechanical and Materials Engineering, University of Nebraska – Lincoln.
2021 – 2022	**Graduate Teaching Assistant – Fulbright Fellow**, Department of Mechanical and Materials Engineering, University of Nebraska – Lincoln.
2018 – 2021	**Lecturer**, Department of Basic Training, Escuela Politécnica Nacional.
2018	**Lecturer**, Department of Industrial Engineering, Universidad Tecnológica Indoamérica, Quito, Ecuador.
2015 – 2017	**Graduate Research Assistant**, Department of Energy, Zhejiang University.
2013 – 2014	**Lecturer**, School of Mechanical Engineering, Escuela Politécnica Nacional.

GRANT & FELLOWSHIPS

Awarded Research Grant

2026	DOE – Genesis: Transforming Science and Energy with AI. "Weak-Form Agentic Digital Twins for Real-Time Control of Magnetically Confined Fusion Systems Under Sparse Observation". Awarded: $750k. Role: **Collaborator**.

Fellowships & Awards - University of Nebraska-Lincoln (Total: $44k)

2025	5.	Doctor Honoris Causa, Humanismus cum Rigoro Scientifico. Awarded by the Universidad GESTALT – Mexico; Ibarra – Ecuador.
2023	4.	Travel Award, College of Engineering, University of Nebraska – Lincoln ($500).
2022	3.	Travel Award, College of Engineering, University of Nebraska – Lincoln ($500).
2021 – 2024	2.	Othmer Fellowship. University of Nebraska – Lincoln's graduate studies for academic merit ($12k).
2021 – 2023	1.	Fulbright Faculty Development Program to study in a Ph.D. program in the United States ($31k).

Fellowships & Awards - Zhejiang University (Total: $19k)

2017	2.	Best Presenter of the session, IEEE CSPA Conference, Penang, Malaysia.
2014 – 2018	1.	Chinese Government Scholarship. Language training and Master's studies.

Fellowships & Awards - Escuela Politécnica Nacional (Total: $7k)

2012 – 2013	2.	Merit-based Teaching Assistantship. Casting Laboratory
2013	1.	Academic Excellence. Top 2% in the Mechanical Engineering program.

PUBLICATIONS

Peer-Reviewed Journals

2026	12.	**López C.**, Partidge M., De Pascuale S., Lore J., Christlieb A.J., Becker S., Bortz D.M., "WSINDy for Model Predictive Control with Applications to Fusion, Drones, and Chaos", *Proceedings of the Royal Society A*.
	11.	Ghosh S., Andrulonis A., **López C.**, Singh A., Moore K.J., "Energy-Based Dual-Phased Dynamics Identification for Forced Nonlinear Vibrations", *Mechanical Systems and Signal Processing*.
	10.	**López C.**, Herzlieb J.E., Moore K.J., "System Identification via Validation and Adaptation for Model Updating Applied to a Nonlinear Cantilever Beam", *ASME Journal of Computational and Nonlinear Dynamics*.
	9.	**López C.**, Moore K.J., "System Identification via Validation and Adaptation (SIVA) for Model Identification of Nonlinear Vibrating Structures", *Nonlinear Dynamics*.
2025	8.	**López C.**, Naranjo Á., Salazar D., Moore K.J., "Weak-form modified sparse identification of nonlinear dynamics", *Journal of Computational Physics*.
	7.	**López C.**, Moore K.J., "Energy-based dual-phase dynamics identification of clearance nonlinearities", *Nonlinear Dynamics*.
	6.	**López C.**, Singh A., Naranjo Á., Moore K.J., "A data-driven, energy-based approach for identifying governing dynamics in vibrating structures directly from measurements", *Mechanical Systems and Signal Processing*.
2024	5.	**López C.**, Moore K.J., "Enhanced adaptive linear chirplet transform for crossing frequency trajectories", *Journal of Sound and Vibration*.
2022	4.	**López C.**, Naranjo A., Lu S., Moore K.J., "Hidden Markov model based stochastic resonance and its application to bearing fault diagnosis", *Journal of Sound and Vibration*.
	3.	**López C.**, Wang D., Naranjo A., Moore K.J., "Box-Cox-sparse-measures-based blind filtering: Understanding the difference between the maximum kurtosis deconvolution and the minimum entropy deconvolution", *Mechanical Systems and Signal Processing*.
2017	2.	**López C.**, Zhong W., Lu S., Cong F., Cortese I., "Stochastic resonance in an underdamped system with FitzHug-Nagumo potential for weak signal detection", *Journal of Sound and Vibration*.

1. Zhong W., Chen X., Zhou Y., Wu Y., **López C.**, "Optimization of a solar aided coal-fired combined heat and power plant based on changeable integrate mode under different solar irradiance", *Solar Energy*.

Preprint

S Ghosh S., **López C.**, Singh A., Moore K.J.,"Decomposition-based Energy-based Dual-Phase Dynamics Identification for Nonlinear MDOF Systems", *Mechanical Systems and Signal Processing*.

CONFERENCE PRESENTATIONS & PROCEEDINGS

* indicates I presented; otherwise, a co-author presented.

2027
17. Ghosh S., **López C.**, Singh A., Moore K.J. "Energy-based Dual-phase Dynamics for Nonlinear System Identification of Multi-Degree-of-Freedom Systems", *International Modal Analysis Conference (IMAC)* XLV, Kissimmee, FL, United States, February 1-4.

2026
16. Ghosh S., **López C.**, Singh A., Moore K.J. "Energy-based Dual-Phase Dynamics Identification For Multiple-Degree-of-Freedom Nonlinear Structures", *NODYCON – The Fifth International Nonlinear Dynamics Conference*, Rome, Italy, September 20–23.
15. **López C.***, Becker S., Bortz D.M. "Data-Driven MPC via WSINDy for Plasma, Drone Control, and Chaotic Systems", *Society for Industrial and Applied Mathematics – Annual Meeting*, Cleveland, OH, United States, July 6–10.
14. **López C.**, Moore K.J. "Data-driven Structural System Identification via Validation and Adaptation", *IMAC XLIV*, Palm Springs, CA, United States, January 19-22.
13. Ghosh S., Andrulonis A., **López C.**, Moore K.J. "A Data-driven Method for System Identification of Forced Nonlinear Vibrations Using Energy Conservation", *IMAC XLIV*, Palm Springs, CA, United States, January 19-22.

2025
12. **López C.**, Moore K.J. "Generative Adversarial Networks with Embedded Physical Knowledge for Structural System Identification", *International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE)*, Anaheim, CA, United States, August 17–20.
11. Ghosh S., Andrulonis A., **López C.**, Moore K.J. "Data-driven Identification of Forced Nonlinear Vibrations using Conservation of Energy", *IDETC/CIE*, Anaheim, CA, United States, August 17–20.
10. **López C.**, Singh A., Naranjo A., Moore K.J. "A Data-Driven Methodology to Identify the Governing Dynamics in Vibrating Structures Through the Analysis of Mechanical Energy", *IMAC XLIII*, Orlando, FL, United States, February 10-13.

2024
9. **López C.***, Singh A., Naranjo A., Moore K.J. "A data-driven methodology to identify the governing dynamics in vibrating structures through the analysis of mechanical energy", *European Nonlinear Oscillations Conference*, Delft, Netherlands, July 22-26.
8. Oliveira F., Johnson B., Singh A., Moore K.J. "Mathematical Modeling of the Nonlinear Dynamics of Bolted Joint Loosening". **I presented my colleagues' work**. *European Nonlinear Oscillations Conference 2024*, Delft, Netherlands, July 22-26.
7. **López C.**, Ramsey T., Moore K.J. "Digital Engineering Test and Modeling Process: Autonomous Methods for Reconciling Test and Model Results", *JANNAF 18th Modeling and Simulation*, Oklahoma City, OK, United States, May 6–9.
6. **López C.***, Moore K.J. "Enhanced Adaptive Linear Chirplet Transform for Multi-Component Signals with Intersecting Instantaneous Frequencies", *IMAC XLII*, Orlando, FL, United States, January 29 – February 1.
5. Arroyo J., Ramsey T., **López C.***, Moore K.J. "Learning Missing Data in Measurements using Meta-Modeling Techniques", *IMAC XLII*, Orlando, FL, United States, January 29 – February 1.

2023
4. **López C.***, Singh A., Naranjo A., Moore K.J. "Data-driven Euler-Lagrange approach for Time Series Analysis", *IMAC XLI*, Austin, TX, United States, February 13–16.

2022	3.	**López C.***, Naranjo A., Moore K.J. "Hidden Markov Model based Stochastic Resonance for bearing fault diagnosis", *IMAC XL*, Orlando, FL, United States, February 7–10.
2017	2.	**López C.***, Zhong W., Cong F., Hidalgo V., "Bearing fault diagnosis based on Ensemble Empirical Mode Decomposition and Teager energy operator", *IEEE 13th International CSPA*, Penang, Malaysia, March 10–12.
2016	1.	**López C.***, Zhong W., Zheng M., "Short-term Electric Load Forecasting Based on Wavelet Neural Network, Particle Swarm Optimization and Ensemble Empirical Mode Decomposition", *Energy Procedia. 8th International Conference on Applied Energy*, Beijing, China, October 8–11.

WORKSHOPS

2026	4.	Scientific Machine Learning. University of Notre Dame, South Bend, IN, USA. September 26–27. Role: **Attendee**.
	3.	Society of Industrial and Applied Mathematics Annual Meeting. Weak-Form Scientific Machine Learning: Theory and Applications Minisymposium. Cleveland, OH, USA. July 6–10. Role: **Co-organizer**.
	2.	WSINDy for model predictive control with applications to plasma boundary dynamics, drone tracking trajectory, and chaotic systems. Postdoctoral seminar, University of Colorado Boulder, Boulder, CO, USA. April 29. Role: **Presenter**.
2025	1.	Center for Hierarchical and Robust Modeling of Non-Equilibrium Transport Annual Meeting. University of Washington, Seattle, WA, USA. December 1–3. Role: **Presenter**.

INDUSTRY EXPERIENCE

2022 – 2023	Cruze Distribution LLC dba ARYSE under AFOSR STTR Phase II, "STTR: Performance Knee Brace". I used finite element models to simulate and experimentally test components of a new type of knee brace.

TEACHING EXPERIENCE & ACTIVITIES

University of Nebraska – Lincoln

Professional Development

2022 – 2023	College of Engineering Graduate Student Teaching Fellow, and Center for the Integration of Research, Teaching, and Learning @ Nebraska Associate Level Certification. I was selected for a one-year STEM teaching program with evidence-based pedagogy and mentored course design.

Course Taught

2021 – 2022	MENG 3250 Mechanics of Elastic Bodies (Teaching Assistant).

Escuela Politécnica Nacional

Courses Taught (in Spanish)

2018 – 2021	- BAS030 Fundamentals of Physics - FIS145 Newtonian Mechanics: Practical Component and FIS151 Laboratory.
2013 – 2014	- MEC1B2 Casting Technology - MEC4E3 Numerical Methods - MEC4B4 Mechanics of Materials - MEC7F4 Welding Production and Maintenance.
2012 – 2013	- MEC1B2 Casting Laboratory (Laboratory Assistant and Auxiliar).

Universidad Tecnológica Indoamérica

Courses Taught (in Spanish)

2018 | - CAD/CAM/CIM and Maintenance.

Mentoring

2026 | 8. Tony Samour, University of Colorado Boulder, Master student.

2025 | 7. Jackson E. Herzlieb, Georgia Institute of Technology, Undergraduate student.

2024 | 6. Alexis Andrulonis, Georgia Institute of Technology, Undergraduate student.

5. Sayantan Ghosh, University of Nebraska – Lincoln, Ph.D. student.

2023 | 4. Thomas Ramsey, University of Nebraska – Lincoln, Undergraduate student.

2022 | 3. Javier Arroyo, University of Nebraska – Lincoln, Ph.D. student.

2014 | 2. Sebastián Camacho. "Design and simulation of a clay modeling lathe", Escuela Politécnica Nacional, Undergraduate student, committee member (in Spanish).

1. Maritza Chuquirima and Lenin Cortez. "Study and Obtaining of Iron Metal from Ferruginous Sands", Escuela Politécnica Nacional, Undergraduate students, committee member (in Spanish).

SERVICE

2024 | 4. Certified Peer Reviewer: Elsevier, Researcher Academy. I completed the modules of the training program.

2023 | 3. Selection Committee – Fulbright Commission in Ecuador: Evaluation of applications.

2022 | 2. Intermediary for Academic Collaboration: Facilitated research partnership between University of Nebraska – Lincoln's Mechanical & Materials Engineering Department and Escuela Politécnica Nacional's Mechanical Engineering School.

2021 – date | 1. Reviewer:

Journals

- Applied Mathematical Modelling
- Chaos, Solitons and Fractals
- Communications Physics
- Discover Applied Science
- IEEE Trans. on Industrial Electronics
- IEEE Trans. on Signal Processing
- Journal of Aircraft
- Structural Health Monitoring
- Journal of Nondestructive Evaluation
- Journal of Signal, Image and Video Proc.
- Journal of Sound and Vibration
- Mechanical Systems and Signal Proc.
- Nonlinear Dynamics
- ROBOMECH Journal
- Scientific Reports

Conference

International Design Engineering Technical Conferences & Computers and Information in Engineering Conference

LANGUAGES

Spanish (native), English (advanced), Mandarin (limited working proficiency).

SOFTWARE PACKAGES

A more detailed list is available at cristianlopez85.github.io/research/.

2026 | 5. WSINDY-MPC, written in MATLAB. A framework that uses WSINDYc and Model Predictive Control to identify interpretable nonlinear dynamical models from noisy measurements and use them for prediction and closed-loop control.

2025 | 4. WmSINDy, written in Python. A system identification method that identifies parsimonious nonlinear dynamical models from noisy measurements while simultaneously characterizing measurement noise.

2025	3. EDDI, written in MATLAB. An energy-based method for the discovery of governing equations directly from free-response measurements.
2024	2. EALCT, written in MATLAB. A signal decomposition method for signals that have crossing instantaneous-frequency trajectories
2017	1. SRFHN, written in MATLAB. A stochastic resonance method using an underdamped FitzHugh–Nagumo potential to enhance weak periodic signals buried in noise.